

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 04 — INTRODUCTION TO SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

Student Edition

Authorized ICDFA training environments only
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Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Procedure

Step 1. Import/start the ICDFA PLC and SCADA VMs using the isolated network configuration supplied by your instructor. Create LAB04-START snapshots if they are not already available.

Step 2. On the SCADA VM, sign in and open a terminal. Start FUXA using the training image command: `sudo fuxa`. Enter the SCADA account password when prompted.

Step 3. Open Firefox and browse to <http://127.0.0.1:1881/editor> to access the local FUXA editor.

Step 4. Use the Save/Open Project function to load TankFillTest.json from the instructor-provided project location.

Step 5. Open FUXA project settings > Connections. Edit the openplc device and set the Modbus TCP endpoint to 192.168.1.10:502. Save the connection.

Step 6. Switch to the PLC VM. Open <http://localhost:8080/> in the local browser and sign in to OpenPLC Runtime using the credentials configured in the training image.

Step 7. Navigate to Programs, select the provided TankFillSim.st program, upload it, name it Tank Fill Sim and allow it to compile.

Step 8. Go to Dashboard and start the PLC. Confirm that the runtime status shows Running.

Step 9. Return to the SCADA VM and FUXA. Run/Play the HMI project.

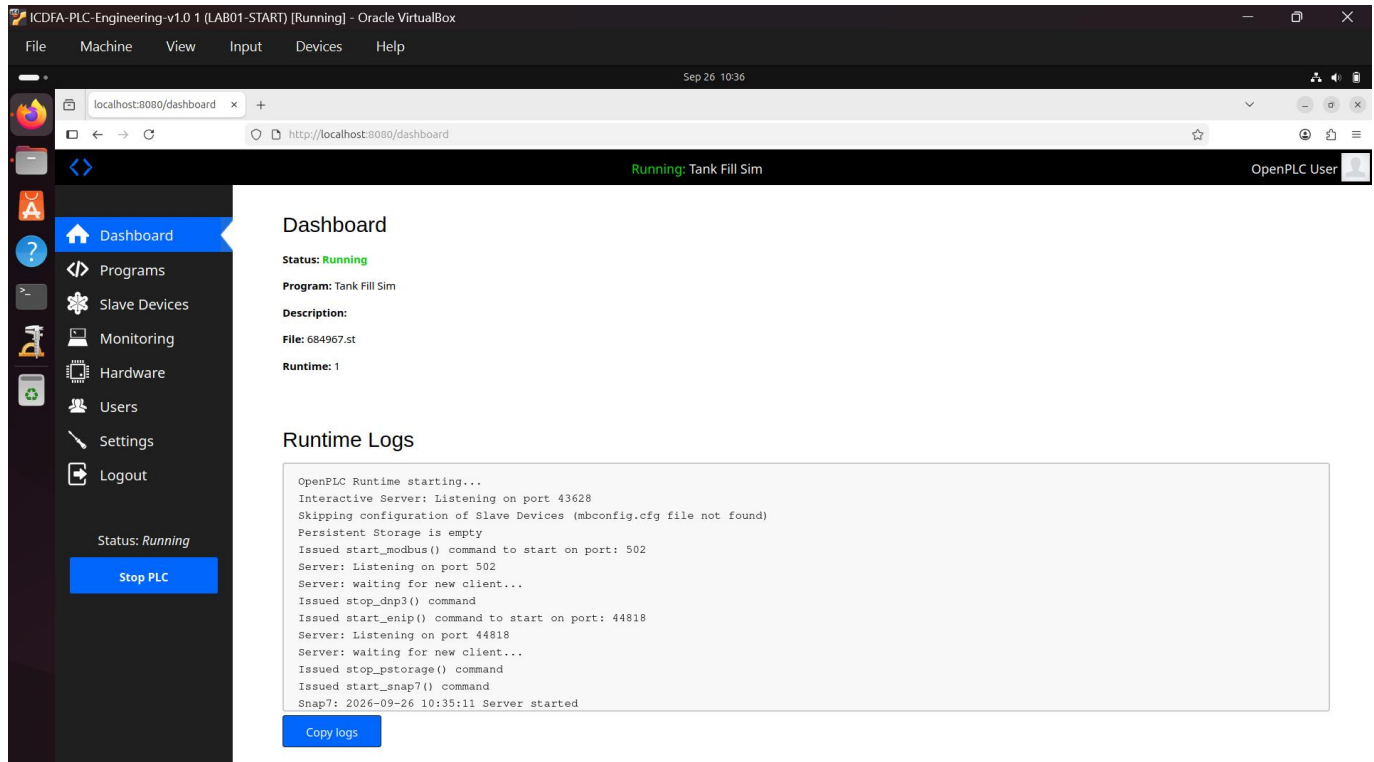
Step 10. Observe the tank-fill simulation. Record the tank-level range, valve open/closed indication and any trend information presented to the operator.

Step 11. Create a simple architecture diagram in your report showing Field/Process -> PLC -> Industrial Protocol/Network -> SCADA/HMI -> Operator, and indicate where alarms/events/historian functions belong conceptually.

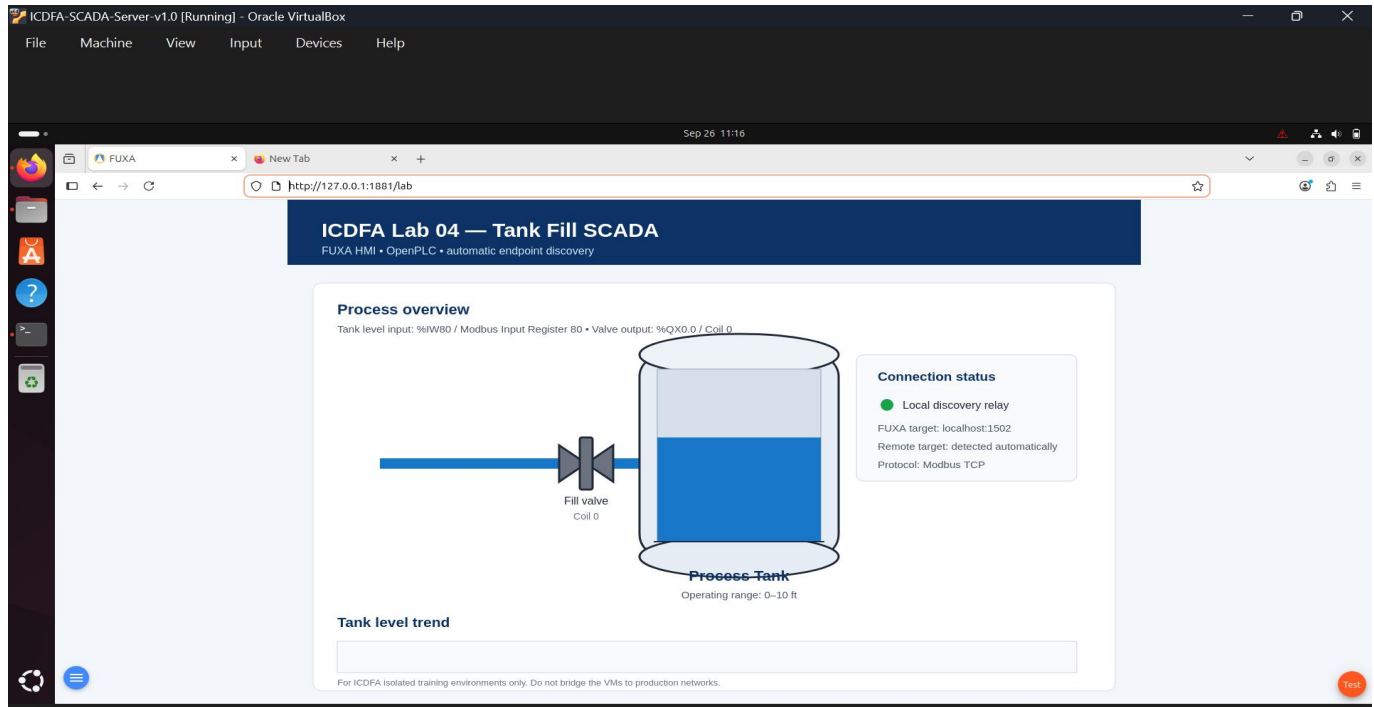
Step 12. Capture the required screenshots and stop the simulation cleanly when finished.

Required Evidence & Submission

- Screenshot showing OpenPLC Runtime in Running state.

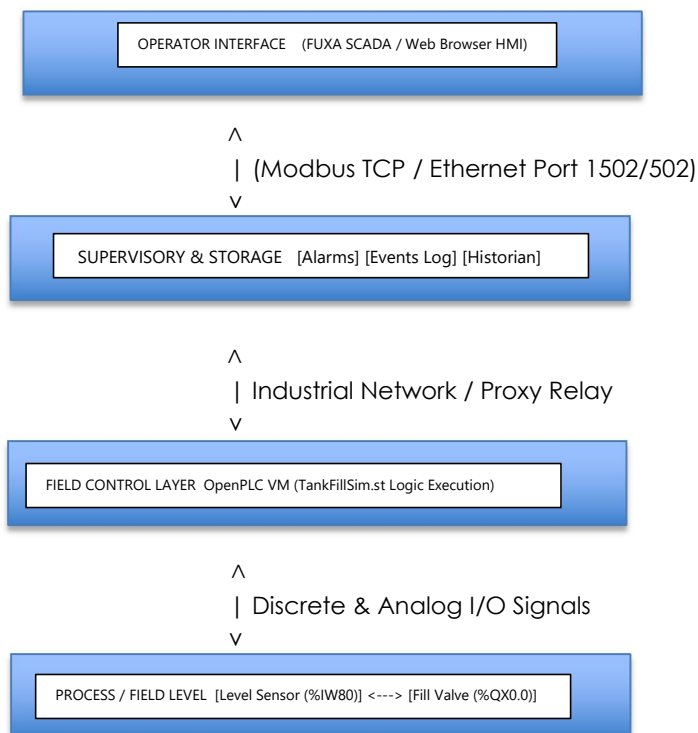


- Screenshot showing the FUXA tank-fill HMI in operation.



- A labelled SCADA architecture diagram.

Answer:



- **A 150–250 word explanation of the difference between control logic in the PLC and supervisory visualization in the HMI/SCADA layer**

Answer:

Control Logic in PLC vs. Supervisory Visualization in SCADA/HMI:

The Programmable Logic Controller (PLC) operates at the field control layer (Purdue Model Level 1), executing real-time, deterministic control code (TankFillSim.st). Its primary function is to process direct input signals from physical or simulated process sensors (such as tank level measurements) and execute control loops to actuate field devices (such as fill valves). The PLC functions independently of human interfaces to ensure constant operational safety and immediate control responses.

In contrast, the Human-Machine Interface (HMI) and SCADA layer operate at Purdue Model Level 2 to provide supervisory oversight, visualization, and human interaction. Rather than making millisecond control decisions, SCADA queries Modbus TCP registers to present human operators with graphical representations (e.g., a 0–10 ft liquid tank gauge), historical trends, alarm management, and event logging. While the PLC autonomously controls valve states based on hardcoded setpoints to prevent tank overflow, the SCADA HMI contextualizes raw process variables into visual indicators, giving operators actionable situational awareness to monitor plant operations safely.

Knowledge Check

- 1. What is the primary purpose of an HMI in a SCADA system?**

Response: To provide human operators with a user-friendly graphical interface to monitor real-time process parameters, observe equipment status, control operational variables, and receive/acknowledge active system alarms.

- 2. What is the difference between an alarm, an event record and historized process data?**

Response: An alarm indicates an abnormal or critical condition requiring immediate operator action; an event record logs operational state changes or operator interactions (e.g., login, mode change) for audit trails, trend analysis and reporting.

- 3. Why should a SCADA operator have context rather than only raw sensor values?**

Response: Raw numerical values lack operational boundaries, health states, and functional impact. Context (such as visual gauges, engineering units, and color-coded safe operating ranges) enables operators to instantly evaluate process safety and make informed decisions.

- 4. What role does the PLC play compared with the SCADA application?**

Response: The PLC handles fast, real-time control logic and physical device I/O at the field layer, whereas the SCADA application aggregates data from controllers to deliver central monitoring, human visualization, alarming, and historical recordkeeping.